Remote Sensing Change Detection via Temporal Feature Interaction and Guided Refinement

Zhenglai Li, Chang Tang, *Member, IEEE*, Lizhe Wang, *Fellow, IEEE*, and Albert Y. Zomaya, *Fellow, IEEE*

***Abstract*— Remote sensing change detection (RSCD), which identifies the changed and unchanged pixels from a registered pair of remote sensing images, has enjoyed remarkable success recently. However, locating changed objects with fine structural details is still a challenging problem in RSCD. In this article, we propose a novel RSCD network via temporal feature interaction and guided refinement (TFI-GR) to solve this issue. Specifically, unlike previous methods, which just employ one single concatenation or subtraction operation for bi-temporal feature fusion, we design a temporal feature interaction module (TFIM) to enhance interaction between bi-temporal features and capture temporal difference information at diverse feature levels. Afterward, a guided refinement modules (GRMs), which aggregates both low- and high-level temporal difference representations to polish the location information of high-level features and filter the background clutters of low-level features, is repeatedly performed. Finally, the multilevel temporal difference features are progressively fused to generate change maps for change detection. To demonstrate the effectiveness of the proposed TFI-GR, comprehensive experiments are performed on three high spatial resolution RSCD datasets. Experimental results indicate that the proposed method is superior to other state-of-the-art change detection methods. The demo code of this work is publicly available at https://github.com/guanyuezhen/TFI-GR.**

***Index Terms*— Change detection, feature pyramid, guided refinement module (GRM), temporal feature interaction module (TFIM).**

I. Introduction

REMOTE sensing change detection aims to provide labels, i.e., changed and unchanged, to each pixel of a registered pair of images that are captured from the same region at different times [1]. This topic has attacked intensive attention owing to its real-world applications, such as global resources monitoring [2], land-use change detecting [3], [4], damage assessment [5], [6], and urban management [7], [8].

In the past few decades, many change detection methods have been proposed. Most of them utilize the handcrafted features as cues to generate change maps, such as change vector analysis [9], Markov random field models [10], and support vector machines [11]. However, the handcrafted features cannot cope with complex real-world scenarios due to their low robustness and insufficient semantics, which significantly limit their performance in remote sensing change detection (RSCD) tasks.

Recently, deep convolutional neural networks (CNNs), which are of great capabilities to represent data in multiple levels of abstractions, have been widely studied and applied in many computer vision tasks, e.g., semantic segmentation [12], [13], object detection [14], [15], and image retrieval [16]. Nowadays, many change detection methods have been proposed to leverage the powerful representation capabilities of CNNs to address the change detection task and shown better performance than traditional approaches [7]. Since the bi-temporal images are collected at different periods, it is difficult to effectively capture the desired change information due to the domain gap. A widely used scheme for deep learning-based methods is to use weight-share Siamese CNNs to extract bi-temporal features and measure temporal changes in the feature space. Based on the way to measure changes, these methods can be generally grouped into two categories, i.e., metric-based ones [17], [18], [19], [20] and classification-based ones [21], [22], [23], [24]. The metric-based methods identify the changes by learning a parameterized embedding distance of paired images, in which the unchanged samples are pushed to be closer, while the changed samples are encouraged to be apart from each other. The classification-based methods detect the changes by classifying the extracted temporal features into different categories, in which the positions of change regions exist with higher scores than these of unchanged ones.

Although previous CD methods have made great progress, it is still a challenging problem for them to locate changed objects with fine structural details as seen in Fig. 1. Some unchanged regions are erroneously detected as changed ones. Besides, the entire change objects cannot be captured. Recent approaches mainly perform bi-temporal feature fusion with single concatenation or subtraction operation [7], [24], [25], [26], [27], [28]. However, just leaving one operation is less

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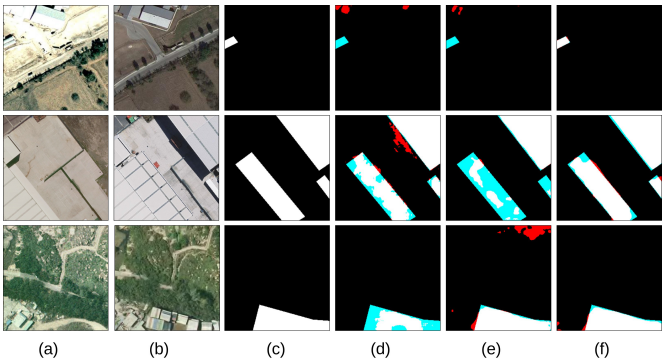


Fig. 1. Visual sample results of diverse methods. (a) T1 images. (b) T2 images. (c) Ground-truth. (d) SNUNet [24]. (e) BIT [27]. (f) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue). These recent proposed methods fail to locate change objects with structural details. Our method obtains high-quality change detection results.

adaptive for capturing the temporal difference information. Apart from bi-temporal feature fusion, multilevel information aggregation also holds a key role in accurate change detection. Generally, the high-level semantic information is beneficial for locating changed objects, while the low-level cues are conducive to identifying the fine-grain details and boundary of changes. Thus, the low-level and high-level features show complementary aspects to each other. Recent approaches mainly integrate multilevel temporal difference features in a coarse-to-fine structure, i.e., U-shape [29], for generating change maps. In [25], two bottom-to-top branches are designed to capture the complementary information among multilevel temporal features. In [24], different levels of features are densely connected from bottom to top to generate a fine-grained change map. However, using a coarse-to-fine structure cannot fully exploit the complementary information of different levels. In addition, the noise information mixed in multiple levels of features inevitably limits the detection performance. The low-level features consist of fines boundary details related to changed objects and background clutters. The high-level features contain location information but coarse boundary related to changed objects. Therefore, the change information may be degraded by the background clutters of low-level features and coarse boundary of high-level ones in a coarse-to-fine fuse structure. Subsequently, they fail to locate changed objects with fine structural details.

To address the above issues, we propose a novel RSCD network by enhancing temporal feature interaction and performing cross-level feature refinement. Specifically, we exploit a siamese CNN to extract multilevel temporal features. Then, we introduce a temporal feature interaction module (TFIM), in which feature concatenation and element-wise subtraction are combined to jointly detect temporal difference information for subsequent processing at diverse feature levels. Afterward, we design a guided refinement module (GRM), which aggregates the change information of pyramid temporal difference features and delivers the aggregated change information to all pyramid features to refine different aspects of changed objects. Finally, the multilevel features are progressively fused to generate a change map for change detection. In summary, the main contributions are as follows.

1) We propose a novel RSCD model, temporal feature interaction, and guided refinement (TFI-GR), which integrates the multilevel temporal difference features to capture rich context information to accurately identify changes.
2) We propose a TFIM, which highlights the changed region of temporal features and generates temporal difference features at diverse feature levels. Our TFIM is capable of capturing valuable temporal difference information and providing detection cues for subsequent processing.
3) We design a GRM, which exploits the cross-level complementary information to refine multilevel features.
4) We conduct a series of experiments to verify the effectiveness and superiority of the proposed method. Experiments on three benchmark datasets prove that our TFI-GR outperforms seven state-of-the-art models in the terms of five evaluation metrics.

The rest of this article is organized as follows. Section II gives a brief review of the most related work. In Section III, we present the details of the proposed method. Section IV provides a series of experimental results, discussion, and model analysis. In Section V, we give a conclusion of this article.

II. Related Work

A. Traditional RSCD Methods

Early RSCD methods usually apply algebra approaches, e.g., image differencing, and image regression [30], [31], [32], [33] to obtain the difference map as features and need to choose a suitable threshold to decide changed regions. Other types of RSCD methods utilize the transform techniques, which enhance the change information by using the means of diverse decompositions, i.e., principal component analysis (PCA) [34] and Tasseled Cap Transformation [35]. Although these methods are easy to implement, they are difficult to recognize changes in complex real-world scenarios due to their low robustness and insufficient semantics shallow features.

B. Deep Learning-Based RSCD Methods

Nowadays, owing to the strong feature representation capability of deep learning, many deep learning-based methods are proposed and achieve excellent results compared with traditional ones. The deep learning-based methods can be generally grouped into two categories, i.e., metric-based ones [17], [18], [19], [20], [36], [37] and classification-based ones [24], [38], [39], [40], [41].

1) Metric-Based Methods: Some RSCD methods learn a parameterized embedding distance of paired bi-temporal images to measure changes. In this parameterized embedding distance space, the unchanged samples are pushed to be closer, while the changed ones are encouraged to be apart from each other. To achieve this goal, an ℓ_1 distance is usually employed to determine whether changes have occurred. Furthermore,

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the contrastive loss [17], [19], [20], and triplet loss [18] are leveraged to constrain the embedding distance in the training process.

2) Classification-Based Methods: Some RSCD approaches detect the changes by classifying the extracted bi-temporal features into different categories, in which the positions of change regions consist of a higher score than these of unchanged ones. With the extracted temporal difference feature representations, classification-based methods usually employ a segmentation head (i.e., a 1×1 convolutional layer) to classify pixels into diverse categories.

Although metric-based methods and classification-based ones measure changes with diverse schemes, their change detection performance is largely dependent on the learned feature representations. To this end, many efforts are devoted to learning discriminative feature representation for the change detection task.

3) Bi-Temporal Information Fusion: Bi-temporal information fusion is a key component in RSCD. Feature concatenation or subtraction operation are two widely used manners to fuse bi-temporal information. Temporal concatenation can retain original bi-temporal information by stacking bi-temporal features along the dimension of channels in order. Feature subtraction measures pixel-wise temporal difference, which may weak change representation ability due to the handcrafted distance, i.e., ℓ_1 or ℓ_2 distance [42]. By considering the time order in temporal concatenation, Zheng *et al.* [42] introduced a temporal-symmetric transformer to learn bi-temporal difference information. Based on feature subtraction, the work in [43] proposes a difference enhancement module (DEM). DEM subtracts bi-temporal features and captures contextual information by a global average pooling operation, which is served as channel attention (CA) to enhance the difference between bi-temporal features.

4) Multilevel Feature Aggregation: The U-shape architecture, e.g., the encoder–decoder structure with nested connections, is widely used in existing RSCD methods with promising results. Daudt *et al.* [21] proposed three different U-net-based models, i.e., FC-early fusion (EF), FC-Siam-conc, and FC-Siam-diff by discussing two image input schemes and two skip connection manners. However, single feature extraction remains difficult to capture complex changes mixed in each skip connection. To this end, UNet++ [22] extracts multiscale features by applying dense skip connections to reduce some pseudo-changes. To fully explore the temporal property of bi-temporal images, the BiDateNet [23] combines skip connection with LSTMs to capture temporally distinguishable features for the change detection task.

However, the temporal features interaction and the discrepancy among temporal difference features of diverse levels have not been comprehensively studied in existing work. How to design effective feature aggregation strategies to generate discriminative difference maps plays a key role in RSCD tasks. Based on the above discussions, we design a change detection network to accurately generate change maps.

III. Proposed Method

In this section, we first present the overall architecture of the proposed method. Then, we introduce the proposed TFIM and GRM. Finally, we provide the hybrid loss fusion.

A. Overall Architecture

As shown in Fig. 2, the proposed method consists of feature extraction, temporal fusion, and change reasoning. For feature extraction, a registered pair of images T1 and T2 are encoded separately though the weight-shared two-stream ResNet-18 [44]. The extracted multilevel temporal features are denoted as $\mathbf{F}_i^1$ and $\mathbf{F}_i^2$, $i \in \{1, 2, 3, 4, 5\}$. Then, we employ the TFIMs to fuse temporal features and capture temporal difference information at multiple feature levels. For change reasoning, we adopt a series of GRMs to aggregate multilevel temporal difference features to extract change information, which is delivered to all pyramid features to polish the details of changed objects. Finally, the change map is obtained by fusion pyramid features progressively.

B. Temporal Feature Interaction Module

Previous change detection methods usually only adopt a feature concatenation operation or an element-wise subtraction operation to obtain the difference information from the bi-temporal features. However, such one operation is less adaptive to change detection tasks. To tackle this dilemma, we propose a TFIM, as shown in Fig. 3. It aims to effectively capture the difference information between the bi-temporal features and provide detection cues for identifying changes. Taking the ith level temporal features as an example, first, we calculate a coarse difference feature map by an element-wise subtraction operation with a subsequent absolution operation as

$$\hat{\mathbf{D}}_i = \mathrm{Conv}_{3\times3}\left(\left|\mathbf{F}_i^1 \ominus \mathbf{F}_i^2\right|\right) \tag{1}$$

where $|\cdot|$ denotes an absolution operation, $\ominus$ is an element-wise subtraction operation, $\hat{\mathbf{D}}_i$ represents a coarse temporal difference feature, $\mathrm{Conv}_{3\times3}(\cdot)$ denotes a 3×3 convolutional layer with a batch normalization and a relu activation function. Considering that $\hat{\mathbf{D}}_i$ can be regarded as element-wise feature weights, we further deliver $\hat{\mathbf{D}}_i$ into temporal features by an element-wise multiplication operation to highlight the changed regions of the temporal features. Afterward, we add the enhanced temporal features with the original ones for refining the feature representations. This procedure can be denoted as

$$\begin{aligned}\hat{\mathbf{F}}_i^1 &= \mathrm{Conv}_{3\times3}\left(\mathbf{F}_i^1 \oplus \mathbf{F}_i^1 \otimes \hat{\mathbf{D}}_i\right)\\ \hat{\mathbf{F}}_i^2 &= \mathrm{Conv}_{3\times3}\left(\mathbf{F}_i^2 \oplus \mathbf{F}_i^2 \otimes \hat{\mathbf{D}}_i\right)\end{aligned} \tag{2}$$

where $\otimes$ and $\oplus$ are element-wise multiply and addition operations, respectively. With the refined bi-temporal features, we concatenate them and pass through a 3×3 convolutional layer to mine the temporal different information and reduce the channels of concatenated feature into half. Then, the temporal difference representation is added with the previous coarse one for refining corresponding difference information. Finally, a 1×1 convolutional layer is used to reduce the channel

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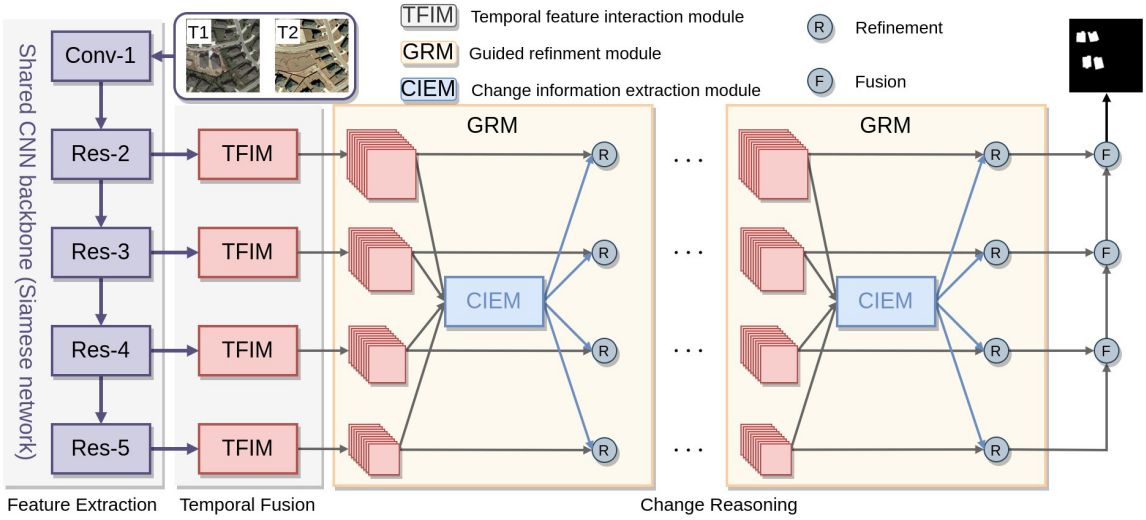


Fig. 2. Framework of the proposed method. The temporal features are extracted from a registered pair of images by a weight-shared ResNet-18 [44]. Then, the temporal features are fused by TFIMs at their corresponding feature levels. The temporal difference features are passed through a series of GRMs to polish the details of changed objects. Finally, a change map is obtained by progressively aggregating temporal difference features.

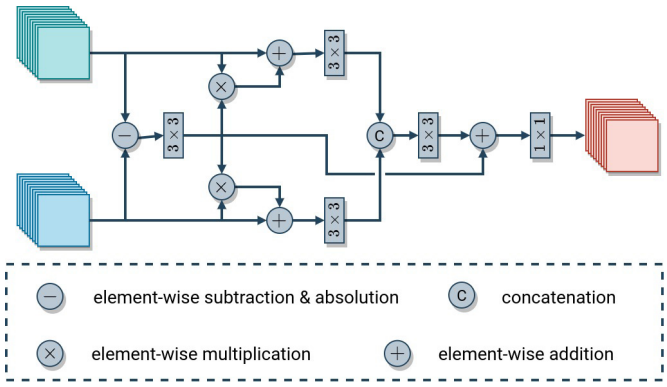


Fig. 3. TFIM.

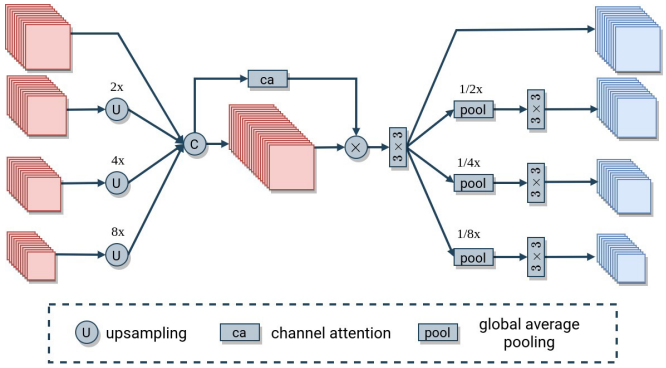


Fig. 4. CIEM.

dimension of the fine temporal difference feature. It can be represented as follows:

$$\mathbf{D}_i = \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Cat}\left(\hat{\mathbf{F}}_i^1, \hat{\mathbf{F}}_i^2\right)\right) \oplus \hat{\mathbf{D}}_i\right) \tag{3}$$

where $\mathrm{Cat}(\cdot)$ is a feature concatenation operation, $\mathbf{D}_i$ denotes the obtained temporal difference feature. We apply the TFIM in four diverse levels of bi-temporal features to generate multilevel temporal difference features to capture rich context information of temporal images.

C. Guided Refinement Module

The multilevel temporal difference features extracted by the proposed TFIMs focus on different aspects of changes. For example, the high-level ones consist of rich semantic information and the low-quality boundary of changed objects. The low-level ones obtain rich details of changes as well as irrelevant background information. For RSCD tasks, the high-level semantic information is beneficial for locating the changed objects, while the low-level visual details help to detect the boundary of the changed object. Thus, it is essential to jointly explore both low-level and high-level information. However, the background clutters of low-level features and low-quality boundary of high-level features inevitably introduce some disturbing information during a coarse-to-fine fuse procedure, resulting in degraded change detection results. To better explore the complementary information of temporal difference features for the change detection tasks, a GRM is presented. As illustrated in Fig. 2, the proposed GRM consists of two sub-modules, i.e., a change information extraction module (CIEM) and a feature refinement module. The CIEM is to aggregate temporal change information from multilevel temporal difference features, while the feature refinement module is designed to filter the background clutters of low-level temporal difference features and correct the location information of high-level ones. Under a series of GRMs, the irrelevant noise information can be removed from multilevel temporal difference features, which are easy to be aggregated for generating change maps in a coarse-to-fine fuse procedure.

1) Change Information Extraction Module: As shown in Fig. 4, we utilize the up-sampling operations to resize

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multilevel temporal difference features into the same resolution, i.e., $[(H/4), (W/4)]$, which H and W are the height and weight of the input image, respectively. Then, we concatenate them to obtain a unified representation, like

$$\hat{\mathbf{D}}^t = \mathrm{Cat}\big(\mathbf{D}_2^t, \mathrm{Up}\big(\mathbf{D}_3^t\big), \mathrm{Up}\big(\mathbf{D}_4^t\big), \mathrm{Up}\big(\mathbf{D}_5^t\big)\big) \tag{4}$$

where $\mathbf{D}_2^t$, $\mathbf{D}_3^t$, $\mathbf{D}_4^t$, $\mathbf{D}_5^t$ denote the multilevel temporal difference features at tth CIEM, $\hat{\mathbf{D}}^t$ is the fused unified representation, $\mathrm{Up}(\cdot)$ is an up-sampling operation.

Then, we utilize the CA module to construct an attention vector $\mathbf{v}^t$ to select more discriminative features. Specifically, we first utilize a channel-wise global average pooling $\mathrm{Gap}_c(\cdot)$ and a channel-wise global max pooling $\mathrm{Gmp}_c(\cdot)$ to obtain two global statistics vectors of the unified representation $\hat{\mathbf{D}}^t$. Afterward, we add two vectors into one, then deliver it into a fully connected layer $\mathrm{fc}(\cdot)$, and finally go through a sigmoid activation function $\delta(\cdot)$ to compute the CA vector $\mathbf{v}^t$

$$\mathbf{v}^t = \delta\big(\mathrm{fc}\big(\mathrm{Gap}_c\big(\hat{\mathbf{D}}^t\big) \oplus \mathrm{Gmp}_c\big(\hat{\mathbf{D}}^t\big)\big)\big). \tag{5}$$

To this end, we exploit the CA vector to model channel correlations and input the channel enhanced feature into a 3×3 convolutional layer to reduce some unimportant channels. The attention process can be formulated as follows:

$$\bar{\mathbf{D}}^t = \mathrm{Conv}_{3\times3}(\hat{\mathbf{D}}^t \otimes \mathbf{v}^t) \tag{6}$$

where $\bar{\mathbf{D}}^t$ is the final obtained unified temporal difference feature at tth CIEM.

After the fusion process, the obtained unified temporal difference feature $\bar{\mathbf{D}}^t$ well encoded the change information. Then, we use the adaptive average pooling to $\bar{\mathbf{D}}^t$ generate pyramid features with diverse scales as

$$\bar{\mathbf{D}}_i^t = \begin{cases} \bar{\mathbf{D}}^t, & \text{if } i = 2 \\ \mathrm{Conv}_{3\times3}(P(\bar{\mathbf{D}}^t)), & \text{otherwise} \end{cases} \tag{7}$$

where $P(\cdot)$ denotes the adaptive average pooling operation.

2) Feature Refinement: As shown in Fig. 2, we employ the extracted change prior features to guide the feature learning process of pyramid features. To achieve this goal, the change information is introduced to multilevel temporal features by element-wise addition operations, like

$$\mathbf{D}_i^{t+1} = \mathrm{Conv}_{3\times3}\big(\mathbf{D}_i^t \oplus \bar{\mathbf{D}}_i^t\big), \quad i = 2, 3, 4, 5 \tag{8}$$

where $\mathbf{D}_i^{t+1}(i = 2, 3, 4, 5)$ denote a multilevel temporal difference features at $t+1$ CIEM. In (8), for low-level temporal difference features, the change prior information is delivered to make them more focused on changed regions so that the clutters of unchanged regions can be removed. For high-level temporal difference features, the change prior information is also used to correct their location information.

During the guided refinement procedure, the change information is introduced to polish the background clutters of low-level features and guide the high-level features to learn some details of changed objects. Considering that a single GRM cannot completely filter the disturbing information of multilevel temporal difference features, here, we employ a series of GRMs to perform feature refinement as shown in Fig. 2. Finally, a coarse-to-fine pathway is used to aggregate multilevel temporal difference features as

$$\mathbf{P}_i = \mathrm{Conv}_{3\times3}\big(\mathbf{D}_i^n + \mathbf{P}_{i+1}\big), \quad i = 2, 3, 4, 5 \tag{9}$$

where $\mathbf{P}_i$ is the ith level aggregate feature, $\mathbf{D}_i^n$ is the refined feature with nth GRM. The change map is obtained from $\mathbf{P}_2$ by using a 1×1 convolutional layer as

$$\mathbf{C} = \mathrm{Conv}_{1\times1}(\mathbf{P}_2). \tag{10}$$

D. Loss Function

For the change detection task, the ratio of changed regions is much less than that of unchanged ones in most cases, which results in a class imbalance problem. To alleviate this issue and guide the network to learn from complex scenes, we adopt a hybrid loss, including a binary cross-entropy loss $\mathcal{L}_{\mathrm{bce}}$ and a dice loss $\mathcal{L}_{\mathrm{dice}}$ [45]. The cross-entropy loss can be formulated as

$$\mathcal{L}_{\mathrm{bce}}(\mathbf{C}, \mathbf{G}) = \mathbf{G} \cdot \log\mathbf{C} + (1 - \mathbf{G}) \cdot \log(1 - \mathbf{C}) \tag{11}$$

where $\cdot$ denotes a dot-product operation, $\mathbf{C}$ and $\mathbf{G}$ are predicted change map and corresponding ground-truth, respectively. The dice loss can be formulated as

$$\mathcal{L}_{\mathrm{dice}}(\mathbf{C}, \mathbf{G}) = 1 - \frac{2 \cdot \mathbf{G} \cdot \mathbf{C}}{\|\mathbf{G}\|_1 + \|\mathbf{C}\|_1} \tag{12}$$

where $\|\cdot\|_1$ denotes the l_1 norm of $\mathbf{C}$. The total loss is represented as

$$\mathcal{L}(\mathbf{C}, \mathbf{G}) = \mathcal{L}_{\mathrm{bce}}(\mathbf{C}, \mathbf{G}) + \mathcal{L}_{\mathrm{dice}}(\mathbf{C}, \mathbf{G}). \tag{13}$$

IV. Experiments

A. Experimental Setup

1) Implementation Details: We implement the proposed method on the Pytorch toolbox and conduct experiments with a single Nvidia Titan V GPU. The Resnet-18 is utilized as our backbone to extract multilevel temporal features. We apply flip and crop to the input images to perform data augmentation. Our model is trained by the Adam optimizer with a momentum of 0.9, the weight decay setting to 0.0001, β_1, and β_2 as 0.9 and 0.99. The learning rate is initially set to 0.0001 and the batch size is 32. The poly learning scheme is applied to adjust the learning rate as $(1 - (\mathrm{cur_iteration}/\mathrm{max_iteration}))^{\mathrm{power}} \times \mathrm{lr}$, where power and max_iteration are set as 0.9 and 20 000, respectively.

2) Evaluation Metrics: We adopt five widely used evaluation metrics [43], [46], [47], namely Kappa coefficient (κ), intersection over union (IoU), $F1$-score ($F1$), recall (Rec), and precision (Pre) to evaluate the performance of RSCD. The detailed definitions of the above evaluation metrics are formulated as follows:

$$\mathrm{Pre} = \frac{\mathrm{Tp}}{\mathrm{Tp} + \mathrm{Fp}} \tag{14}$$

$$\mathrm{Rec} = \frac{\mathrm{Tp}}{\mathrm{Tp} + \mathrm{Fn}} \tag{15}$$

$$F1 = \frac{2\mathrm{Pre} \cdot \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}} \tag{16}$$

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$$\mathrm{OA} = \frac{\mathrm{Tp} + \mathrm{Tn}}{\mathrm{Tp} + \mathrm{Fp} + \mathrm{Tn} + \mathrm{Fn}} \tag{17}$$

$$\mathrm{P}_e = \frac{(\mathrm{Tn} + \mathrm{Fn}) \cdot (\mathrm{Tn} + \mathrm{Fp}) + (\mathrm{Fp} + \mathrm{Tp}) \cdot (\mathrm{Fn} + \mathrm{Tp})}{(\mathrm{Tp} + \mathrm{Fp} + \mathrm{Tn} + \mathrm{Fn})^2} \tag{18}$$

$$\kappa = \frac{\mathrm{OA} - \mathrm{P}_e}{1 - \mathrm{P}_e} \tag{19}$$

$$\mathrm{IoU} = \frac{\mathrm{Tp}}{\mathrm{Tp} + \mathrm{Fp}} \tag{20}$$

where Tp, Fp, Tn, Fn denote the numbers of true positives, false positives, true negatives, and false negatives, respectively.

B. Datasets

In our experiments, we use three RSCD benchmark datasets. The detailed information is listed as follows.

1) LEVIR [7]: This building change detection dataset exists of 637 pairs of bi-temporal remote sensing images with 1024 × 1024 spatial size and 0.5-m spatial resolution, which is obtained from the Google Earth platform. Under its official dataset split setting, we none overlap crop images into 256 × 256 size of patches and obtained a total of 7120/1024/2048 image pairs for training/validation/test, respectively.

2) SYSU [25]: This dataset consists of 20 000 pairs of bi-temporal remote sensing image patches with 256 × 256 spatial size and 0.5-m spatial resolution. The ratios of training, validation, and test are officially set to 6:2:2. This dataset contains various types of complex change scenes, including road expansion, newly built urban buildings, change of vegetation, suburban dilation, and groundwork before construction.

3) BCDD[1]: This is a building change detection dataset that contains a pair of images with a size of 32 507 × 15 354 and a high resolution of 0.075 m. We first none overlap crop images into 256 × 256 size of patches, and then obtained a total of 6095/762/763 image pairs for training/validation/test as [27], respectively.

C. Ablation Studies

There are two key modules in the proposed method, i.e., TFIM and guide refinement module. Accordingly, we conduct ablation studies on three RSCD datasets to verify the effectiveness of these two modules.

1) Effectiveness of TFIM: The TFIM aims to enhance the information interaction between temporal features, highlight the changed regions, and produce better difference cues for change detection. To validate its effectiveness, we introduce two baseline methods. For the first one, we remove other processes in TFIM and only use the element-wise subtraction and absolution operations to measure difference information at various feature levels, like

$$\mathbf{D}_i = \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{3\times3}\left(\left|\mathbf{F}_i^1 \ominus \mathbf{F}_i^2\right|\right)\right). \tag{21}$$

For the rest one, we remove other processes in TFIM and only use the feature concatenation operations to fuse temporal features at various feature levels, like

$$\mathbf{D}_i = \mathrm{Conv}_{1\times1}\left(\mathrm{Conv}_{3\times3}\left(\mathrm{Cat}\left(\mathbf{F}_i^1, \mathbf{F}_i^2\right)\right)\right). \tag{22}$$

[1]http://study.rsgis.whu.edu.cn/pages/download/building_dataset.html

TABLE I

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS IN TERMS OF κ, IOU, $F1$, OA, REC, AND PRE ON LEVIR, BCDD, AND SYSU DATASETS

	Temporal Fusion	# GRM	κ	IoU	F1	Rec	Pre
LEVIR	Concatenation	0	0.8936	0.8165	0.8990	0.9007	0.8973
	Subtraction	0	0.8919	0.8140	0.8974	0.8949	0.9000
	PCS	0	0.8840	0.8018	0.8900	0.8979	0.8822
	DEM	0	0.8932	0.8158	0.8985	0.8867	0.9107
	TFIM	0	0.8973	0.8222	0.9024	0.8860	0.9195
	TFIM	1	0.9002	0.8268	0.9052	0.8919	0.9188
	TFIM	2	0.9045	0.8336	0.9093	0.8973	0.9215
	TFIM	3	0.9037	0.8323	0.9084	0.8861	0.9319
	TFIM	4	0.9029	0.8311	0.9077	0.8864	0.9301
BCDD	Concatenation	0	0.9364	0.8852	0.9391	0.9234	0.9554
	Subtraction	0	0.9364	0.8852	0.9391	0.9335	0.9447
	PCS	0	0.9336	0.8804	0.9364	0.9187	0.9548
	DEM	0	0.9342	0.8814	0.9370	0.9241	0.9503
	TFIM	0	0.9385	0.8887	0.9411	0.9357	0.9465
	TFIM	1	0.9402	0.8916	0.9427	0.9321	0.9536
	TFIM	2	0.9399	0.8912	0.9425	0.9317	0.9535
	TFIM	3	0.9306	0.8753	0.9335	0.9134	0.9545
	TFIM	4	0.9392	0.8899	0.9417	0.9327	0.9509
SYSU	Concatenation	0	0.7521	0.6770	0.8074	0.7664	0.8531
	Subtraction	0	0.7789	0.7083	0.8292	0.8010	0.8594
	PCS	0	0.7446	0.6685	0.8013	0.7565	0.8517
	DEM	0	0.7877	0.7200	0.8372	0.8286	0.8461
	TFIM	0	0.7840	0.7131	0.8326	0.7948	0.8741
	TFIM	1	0.7805	0.7127	0.8323	0.8318	0.8327
	TFIM	2	0.7902	0.7240	0.8399	0.8437	0.8361
	TFIM	3	0.7856	0.7193	0.8367	0.8462	0.8274
	TFIM	4	0.7782	0.7069	0.8283	0.7942	0.8654

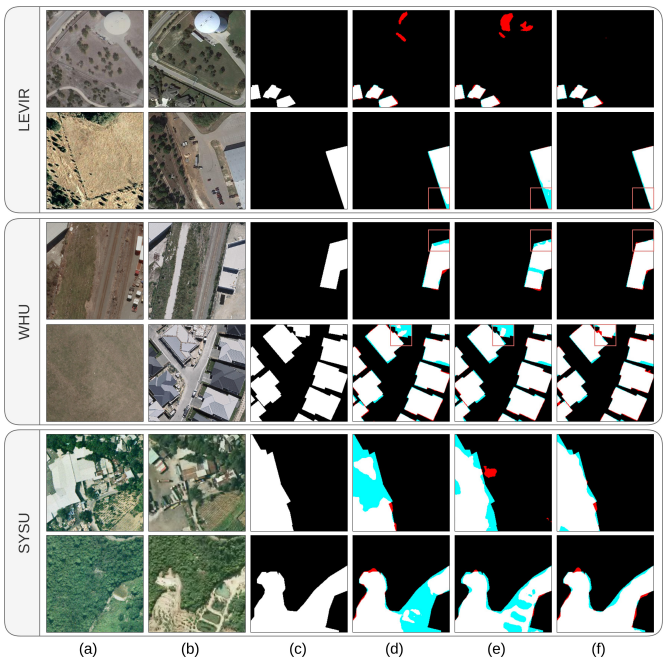


Fig. 5. Visual comparisons of the proposed method with different temporal fusion manners. (a) T1 images. (b) T2 images. (c) Ground-truth. (d) Change detection results of the proposed method with element-wise subtraction operation. (e) Change detection of the proposed method with feature concatenation. (f) Change detection results of the proposed method with TFIM. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

The change detection results in terms of five evaluation metrics on LEVIR, BCDD, and SYSU datasets are reported in Table I, respectively. As can be seen from the results,

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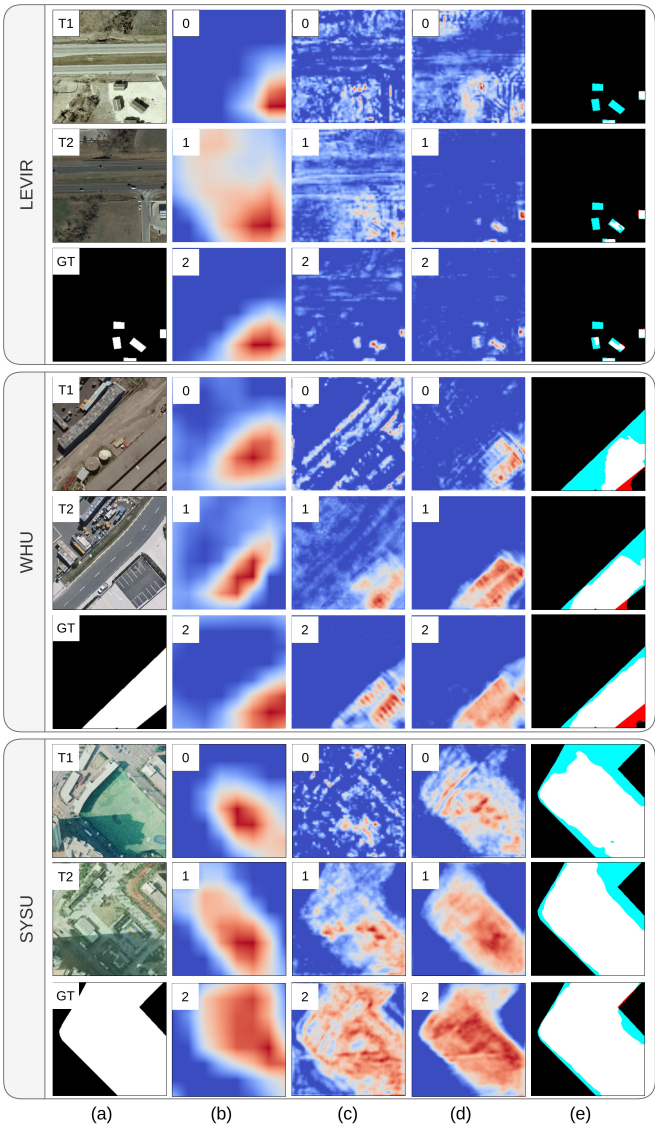


Fig. 6. Visual comparisons of the proposed method with different times of GRMs. (a) T1 images, T2 images, and ground-truth. (b) High-level temporal difference features with diverse times of GRMs. (c) Low-level temporal difference features with diverse times of GRMs. (d) Fused temporal difference features with diverse times of GRMs. (e) Change detection results of the proposed method with diverse times of GRMs. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

when we utilize the proposed TFIM to perform temporal fusion instead of just using concatenation or subtraction, the proposed method can obtain better change detection performance. As shown in Fig. 5, when we only employ the feature concatenation or element-wise subtraction to conduct temporal fusion, the change results are degraded by the background noises, leading to some mistaken predictions (i.e., the first row of LEVIR dataset), and missing prediction (i.e., the second row of SYSU dataset). This indicates that the TFIM can better capture temporal difference information among temporal features.

To further verify the effectiveness of the proposed TFIM, we compare it with two recently introduced modules, including parallel convolutional structure (PCS) [48] and DEM [43]. PCS first concatenate bi-temporal features into one, then utilizes four separate group convolutions with a kernel size of 3×3 and different dilation rate to capture multiscale information. For DEM, it obtains temporal difference representation with a subtraction operation and an absolute value operation. Then, DEM formulates a CA vector and uses it to enhance bi-temporal features. Finally, DEM constructs temporal difference representation from the enhanced bi-temporal features. We plug PCS and DEM into our network and retrain them with the same settings. The change detection results measured by κ, IoU, $F1$, OA, Rec, and Pre on three RSCD datasets are reported in Table I. As can be seen, our proposed TFIM consistently achieves better results than PCS and DEM.

2) Effectiveness of GRM: In the proposed method, we employ the GRM to guide the learning process of pyramid features. To verify the effectiveness of GRM, we report the detection performance evaluated with five metrics by performing different numbers of GRM in Table I. From the results in Table I, just applying one GRM does not obtain the best performance, since one GRM has limited capability in guiding the learning process of multilevel temporal difference features. Our method with two GRMs outperforms the baseline, i.e., 1.14, 0.25, and 1.09 percentage improvements in terms of IoU in LEVIR, BCDD, and SYSU datasets. When the number of GRMs is bigger than 2, the performance decreases compared with two GRMs. The reason may be that adding too many GRMs leads to over-fitting. Thus, considering both the performance and efficiency synthetically, we use two GRMs to polish multilevel temporal difference features in our experiments.

We present the feature maps and predicted change maps of the proposed method with different numbers of GRMs in Fig. 6. From the results, we can observe that, with the number of GRMs increasing, the background clutters of low-level temporal difference features can be removed, meanwhile, the location information of high-level ones becomes more accurate. As a result, the fused features show better boundary and details of change objects and can provide more accurate change detection results. These observations strongly indicate the effectiveness of GRM.

D. Comparison With the State-of-the-Art

We compare the proposed model with seven state-of-the-art RSCD approaches, including four CNNs-based ones: FC-Diff [21], CDNet [49], STANet [7], SRCD-Net [26], MSPSNet [48], SNUNet [24], and a transformer-based one: BIT [27]. We reproduce them by using their released code under their default parameters for fair comparisons.

1) Quantitative Evaluation: Table II reports the quantitative validation comparisons of different methods in terms of κ, IoU, F1, Rec, and Pre on three RSCD datasets. As seen from Table II, the proposed method significantly performs better than existing methods, such as the proposed method improves 6.49, 7.41, and 5.21 percent on the SYSU dataset in terms

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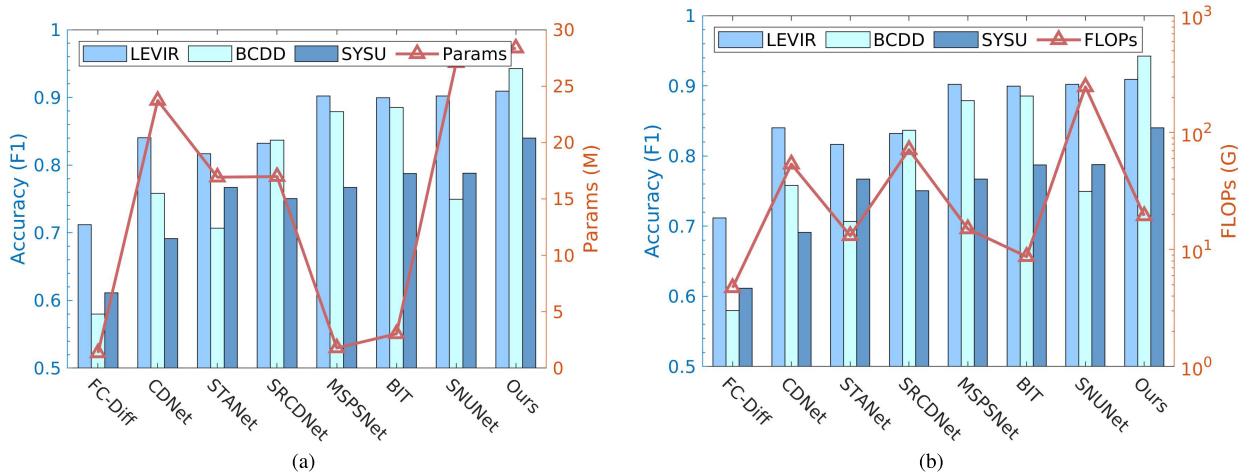


Fig. 7. Model complexity comparison of different methods in terms of parameters (memory cost), FLOPs (computational cost), and $F1$-score on LEVIR, BCDD, and SYSU datasets, respectively. (a) Accuracy versus parameters. (b) Accuracy versus FLOPs.

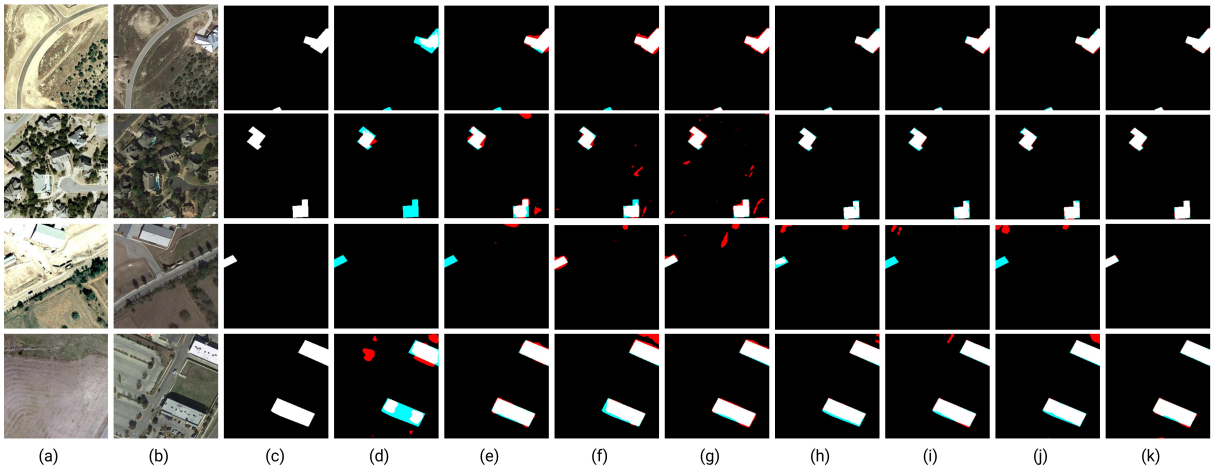


Fig. 8. Visual comparisons of the proposed method and the state-of-the-art approaches on LEVIR dataset. (a) T1 images. (b) T2 images. (c) Ground-truth. (d) FC-Diff. (e) CDNet. (f) STANet. (g) SRCDNet. (h) MSPSNet. (i) BIT. (j) SNUNet. (k) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

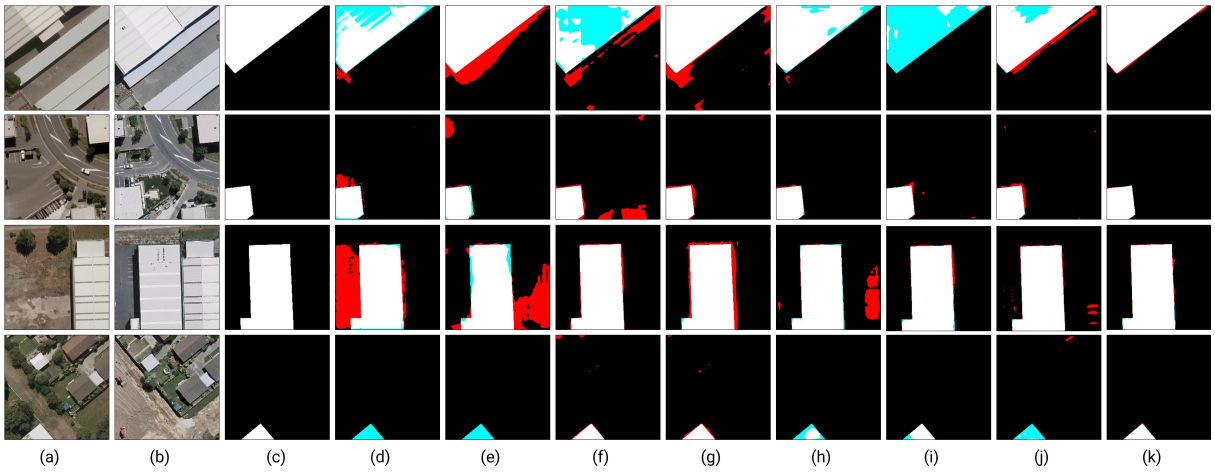


Fig. 9. Visual comparisons of the proposed method and the state-of-the-art approaches on BCDD dataset. (a) T1 images. (b) T2 images. (c) Ground-truth. (d) FC-Diff. (e) CDNet. (f) STANet. (g) SRCDNet. (h) MSPSNet. (i) BIT. (j) SNUNet. (k) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

of κ, IoU, and $F1$, respectively. The improvement can be consistently observed on other datasets and other evaluation metrics. All these measures strongly indicate the effectiveness and superiority of the proposed method.

Furthermore, we adopt two metrics, including the number of model parameters (Params) and the number of floating-point operations (FLOPs), to measure space complexity and computational costs of the networks, respectively. Note that the value

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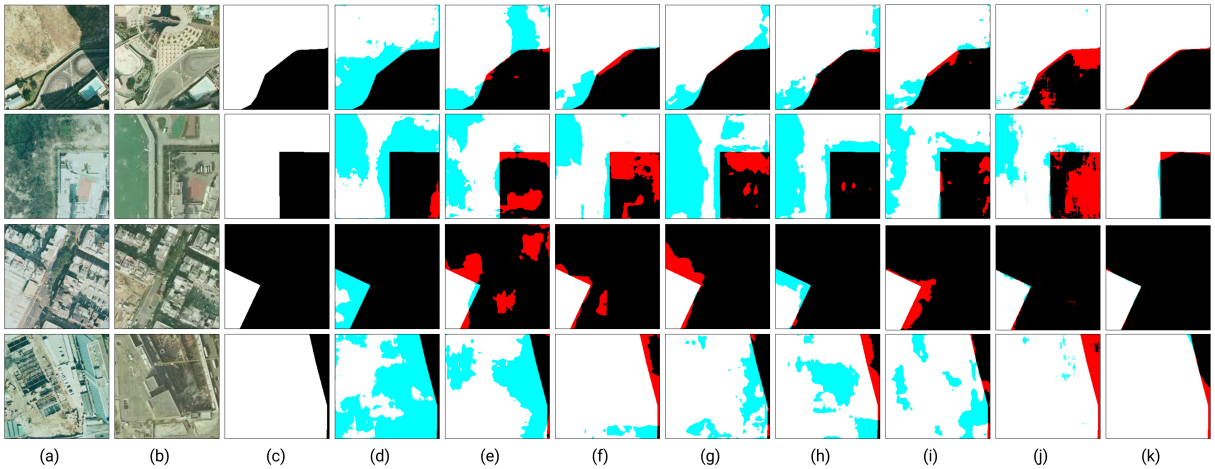


Fig. 10. Visual comparisons of the proposed method and the state-of-the-art approaches on SYSU dataset. (a) T1 images. (b) T2 images. (c) Ground-truth. (d) FC-Diff. (e) CDNet. (f) STANet. (g) SRCDNet. (h) MSPSNet. (i) BIT. (j) SNUNet. (k) Ours. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

TABLE II

QUANTITATIVE COMPARISONS IN TERMS OF κ, IOU, $F1$, OA, REC, AND PRE ON THREE RSCD DATASETS. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Datasets	Methods	Kappa	IoU	F1	Recall	Precision
LEVIR	FC-Diff	0.6992	0.5527	0.7119	0.6070	0.8606
	CDNet	0.8309	0.7246	0.8403	0.9280	0.7678
	STANet	0.8065	0.6902	0.8167	0.8444	0.7907
	SRCDNet	0.8217	0.7123	0.8320	0.9558	0.7366
	MSPSNet	0.8967	0.8213	0.9019	0.8952	0.9087
	BIT	0.8940	0.8172	0.8994	0.8956	0.9033
	SNUNet	0.8970	0.8218	0.9022	0.8929	0.9117
	Ours	0.9045	0.8336	0.9093	0.8973	0.9215
BCDD	FC-Diff	0.5547	0.4081	0.5797	0.8487	0.4402
	CDNet	0.7461	0.6107	0.7583	0.8661	0.6744
	STANet	0.6912	0.5465	0.7067	0.8563	0.6016
	SRCDNet	0.8284	0.7192	0.8367	0.9590	0.7420
	MSPSNet	0.8736	0.7839	0.8789	0.8553	0.9038
	BIT	0.8800	0.7940	0.8852	0.8986	0.8722
	SNUNet	0.7365	0.5996	0.7497	0.9116	0.6367
	Ours	0.9399	0.8912	0.9425	0.9317	0.9535
SYSU	FC-Diff	0.5365	0.4404	0.6115	0.4654	0.8912
	CDNet	0.5859	0.5281	0.6912	0.7496	0.6412
	STANet	0.6923	0.6221	0.7670	0.7907	0.7447
	SRCDNet	0.6752	0.6006	0.7505	0.7384	0.7630
	MSPSNet	0.6994	0.6223	0.7672	0.7348	0.8025
	BIT	0.7243	0.6494	0.7874	0.7648	0.8114
	SNUNet	0.7253	0.6499	0.7878	0.7613	0.8162
	Ours	0.7902	0.7240	0.8399	0.8437	0.8361

of FLOPs highly indicates energy consumption for the network inference, while the number of Params is related to storage consumption for the model. The model complexity in terms of Params and FLOPs are given in Fig. 7. From the results, although the proposed method costs slightly larger parameters, it speeds relatively lower computational cost to obtain better change detection results than other compared methods.

2) Qualitative Evaluation: Figs. 8–10 display the visual comparisons of different methods on three RSCD datasets. For better visualization, we adopt several colors to represent true positives (white), false positives (red), true negatives (black), and false negatives (blue). From the results, we observe that the proposed method exhibits superiority in the following aspects.

1) *Advantages in Location Accuracy:* Compared with other methods, the proposed method can more accurately locate the changed objects and performs well in suppressing the interference of background in bi-temporal images. In the second row of Fig. 8, CDNet, STANet, and SRCDNet can hardly remove irrelevant background clutters, resulting in the wrong prediction. By contrast, the changed buildings are accurately and completely identified by the proposed method.
2) *Advantages in Distinguishing Irrelevant Changes:* In the fourth row of Fig. 8, the same buildings show different colors at different times which causes some irrelevant changes contrary to the real building changes. Many methods (i.e., CDNet, SRCDNet, MSPNet, BIT, and SNUNet) fail to exclude these changes, while the proposed method can well handle the irrelevant changes resulting from seasonal changes or land cover appearance diversity. In the first to third rows of Fig. 9, many methods cannot distinguish the irrelevant changes which contain a similar appearance to buildings. On the contrary, the proposed method can well identify the boundary and body of the building changes.
3) *Advantages in Content Integrity:* In Fig. 10, the complex sense changes cannot be entirely identified by these compared methods. The proposed method can achieve relatively intact detection results. The main reasons are that, first, the proposed method performs temporal feature interaction at multiple levels and generates multilevel temporal difference feature representations which capture the rich contextual information at diverse levels. Second, the proposed method progressively refines multilevel temporal difference features to provide a finer change map for RSCD tasks.

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V. Conclusion

In this work, we proposed a novel RSCD network via TFI-GR. First, the proposed method employs a Siamese CNN to extract multilevel features of the temporal images. Then, it further enhances the temporal feature interaction and highlights the changed regions of features at diverse feature levels. Besides, it refines multilevel temporal difference features by a series of GRMs, which deliver change information to guide the learning process of pyramid features. Finally, the change map is obtained by progressive aggregating multilevel temporal difference features in a coarse-to-fine scheme. Experimental results obtained on three high spatial resolution RSCD datasets indicate that the proposed method is superior to the state-of-the-art change detection methods.

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References

[1] A. Singh, "Review article digital change detection techniques using remotely-sensed data," *Int. J. Remote Sens.*, vol. 10, no. 6, pp. 989–1003, 1989.

[2] R. E. Kennedy *et al.*, "Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects," *Remote Sens. Environ.*, vol. 113, no. 7, pp. 1382–1396, 2009.

[3] J. S. Deng, K. Wang, Y. H. Deng, and G. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.

[4] Y. Hu *et al.*, "An automatic approach for land-change detection and land updates based on integrated NDVI timing analysis and the CVAPS method with GEE support," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 347–359, Dec. 2018.

[5] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007.

[6] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[7] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[8] H. Liu, M. Yang, J. Chen, J. Hou, and M. Deng, "Line-constrained shape feature for building change detection in VHR remote sensing imagery," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 10, p. 410, 2018.

[9] R. D. Johnson and E. S. Kasischke, "Change vector analysis: A technique for the multispectral monitoring of land cover and condition," *Int. J. Remote Sens.*, vol. 19, no. 3, pp. 411–426, 1998.

[10] T. Kasetkasem and P. K. Varshney, "An image change detection algorithm based on Markov random field models," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 8, pp. 1815–1823, Aug. 2002.

[11] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogram. Remote Sens.*, vol. 61, no. 2, pp. 125–133, 2006.

[12] Z. Jin *et al.*, "Mining contextual information beyond image for semantic segmentation," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 7231–7241.

[13] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 4096–4105.

[14] Y. Cui, L. Yan, Z. Cao, and D. Liu, "TF-blender: Temporal feature blender for video object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8138–8147.

[15] Z. Zheng *et al.*, "HyNet: hyper-scale object detection network framework for multiple spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 1–14, Aug. 2020.

[16] S. Kim, D. Kim, M. Cho, and S. Kwak, "Proxy anchor loss for deep metric learning," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 3238–3247.

[17] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep Siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Aug. 2017.

[18] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2018.

[19] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Dec. 2016.

[20] M. Wang, K. Tan, X. Jia, X. Wang, and Y. Chen, "A deep Siamese network with hybrid convolutional feature extraction module for change detection based on multi-sensor remote sensing images," *Remote Sens.*, vol. 12, no. 2, p. 205, 2020.

[21] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[22] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, 2019.

[23] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 214–217.

[24] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[25] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[26] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-Resolution-Based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.

[27] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[28] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[29] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput.-Assist. Intervent.* Springer, 2015, pp. 234–241.

[30] P. R. Coppin and M. E. Bauer, "Digital change detection in forest ecosystems with remote sensing imagery," *Remote Sens. Rev.*, vol. 13, nos. 3–4, pp. 207–234, 1996.

[31] Y. Sun, L. Lei, X. Tan, D. Guan, J. Wu, and G. Kuang, "Structured graph based image regression for unsupervised multimodal change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 16–31, Mar. 2022.

[32] Y. Sun, L. Lei, D. Guan, M. Li, and G. Kuang, "Sparse-constrained adaptive structure consistency-based unsupervised image regression for heterogeneous remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[33] Y. Sun, L. Lei, X. Li, X. Tan, and G. Kuang, "Structure consistency-based graph for unsupervised change detection with homogeneous and heterogeneous remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–21, 2021.

[34] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, 2009.

[35] E. P. Crist, "A TM tasseled cap equivalent transformation for reflectance factor data," *Remote Sens. Environ.*, vol. 17, no. 3, pp. 301–306, 1985.

[36] W. Zhao, L. Mou, J. Chen, Y. Bo, and W. J. Emery, "Incorporating metric learning and adversarial network for seasonal invariant change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2720–2731, Apr. 2019.

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[37] X. Tang *et al.*, "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.
[38] K. Yang *et al.*, "Asymmetric Siamese networks for semantic change detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2021.
[39] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2020.
[40] L. Ru, B. Du, and C. Wu, "Multi-temporal scene classification and scene change detection with correlation based fusion," *IEEE Trans. Image Process.*, vol. 30, pp. 1382–1394, 2020.
[41] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 15193–15202.
[42] Z. Zheng, Y. Zhong, S. Tian, A. Ma, and L. Zhang, "ChangeMask: Deep multi-task encoder-transformer-decoder architecture for semantic change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 183, pp. 228–239, Jan. 2022.
[43] T. Lei *et al.*, "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022.
[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.
[45] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[46] L. Ding, H. Guo, S. Liu, L. Mou, J. Zhang, and L. Bruzzone, "Bi-temporal semantic reasoning for the semantic change detection in HR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[47] J. Huang, Q. Shen, M. Wang, and M. Yang, "Multiple attention Siamese network for high-resolution image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.
[48] Q. Guo, J. Zhang, S. Zhu, C. Zhong, and Y. Zhang, "Deep multiscale Siamese network with parallel convolutional structure and self-attention for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.
[49] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.



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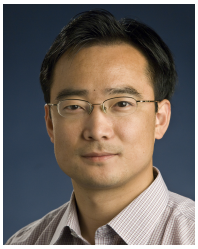
His research interests include multiview clustering and remote sensing change detection.



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